17SLA1

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Problem

Let $a_1, a_2, \ldots a_n, k$, and M be positive integers such that

$$\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n} = k \quad \text{and} \quad a_1 a_2 \cdots a_n = M.$$

If $M > 1$, prove that the polynomial

$$P(x) = M(x+1)^k - (x+a_1)(x+a_2)\cdots(x+a_n)$$

has no positive roots.

Solution. Due to AM-GM,

$$x + a_1 = (x+1) + \underbrace{1 + 1 + \cdots + 1}_{a_1 - 1 \text{ times}} \geq a_1(x+1)^{\frac{1}{a_1}}$$

As a result $(x+a_1)\cdots(x+a_n) \geq M(x+1)^k$, equality iff $a_1 = a_2 = \cdots = a_n$. As $M > 1$, $\exists a_i > 1$. As a result equality will never hold and $(x+a_1)\cdots(x+a_n) \geq M(x+1)^k$ holds for all positive real x.